

Steven McDonald

Senior Frontend Engineer · Tech Lead

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Software engineer leading the frontend of a clinical data platform serving 4,100+ daily users at VA hospitals and private medical groups as it grew from 3 to 15 customer sites. Previously managed an international team of 3 engineers.

EXPERIENCE

HealthLevel — Mountain View, CA · Healthcare data management SaaS · 50 employees

Head of Data Visualization (Aug 2026 – Present) · UI/UX Technical Lead (Oct 2023 – Present) · Technical Manager (Aug 2022 – Oct 2023) · Software Engineer (Sep 2021 – Aug 2022)

Architecture & Platform

- **Proposed and led migration of a 160K-line application** from Backbone/Mithril/jQuery to React and TypeScript, designing a strangler-pattern integration layer that rendered React components inside the legacy framework. No feature freeze, no blocked releases. Cut feature turnaround from ~2 months to ~2 weeks and new-engineer ramp-up from ~6 months to under 2.
- **Established frontend architecture standards** of Redux Toolkit/RTK Query state management, strict TypeScript, typed API contracts, and Jest/React Testing Library coverage, eliminating recurring integration failures.
- **Wrote the core React component library** used by all 16 engineers on the product team, enforcing consistent design and behavior across the platform.
- **Cut page load time (first contentful paint) from 30s to 0.5s** and raised the Lighthouse performance score from 32 to 86 through route-level lazy loading, file compression, profiler-driven re-render reduction, and cache tuning; reduced Jenkins CI/CD build time from 15 to 10 minutes.

Product

- **Eliminated an 80+ hour customer-onboarding bottleneck** by architecting a schema-driven forms engine, form builder, and data-mapping workflow.
- **Owned end-to-end delivery of a telestroke data portal** for the National TeleStroke Program. Ran physician interviews, iterated Figma prototypes, and shipped in 4 months. The portal now processes ~100 stroke cases/month, reduced physician errors 25% and time-per-case 20%, and led directly to a second customer signing.
- **Redesigned core clinical workflows to reduce user fatigue**, cutting required clicks 50% and API calls 30%, and building the theming architecture for a dark mode now used daily by radiologists on overnight shifts.
- **Own the platform's data-visualization strategy**, modernized Highcharts dashboards, and built configurable, reusable visualization patterns.
- **Made the full frontend accessible to users with disabilities**, leading WCAG 2.1 AA remediation ahead of a federal Section 508 audit required for VA contracts; raised the Lighthouse accessibility score from 55 to 92 with accessible primitives verified in axe DevTools and VoiceOver.

Leadership

- **Led technical direction for 9 engineers and 1 designer**; previously managed 3 direct reports across US and India time zones with zero attrition over two years. Instituted the design-to-code pipeline: Figma design tokens compiled to CSS, PM-approved mockups for every feature, and design verification before sign-off.

On Me Gifting — San Francisco, CA · Digital gifting platform startup

Consultant - Aug 2023 – Oct 2023

- Built the first generation of the company's UI framework in Next.js and Tailwind CSS and shipped cross-platform iOS/Android features in React Native.

PROJECTS

anycourseai.com - Apr 2026 – Present

- Schema-driven engine that generates and runs LLM-authored courses (16 published), with a RAG pipeline grounding content in college-level sources.

the2senses.com - Mar – June 2021

- Next.js color-blindness and hearing test suite for rural Indian schools without medical access; deployed by NGOs to screen 500+ students. Winner, "Tech for Good," FDM Technology Awards 2022.

EDUCATION

Cornell University — Ithaca, NY · B.S. Computer Science, cum laude Dec 2019

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML5, CSS3, SASS

Frontend: React, Next.js, React Native, Redux Toolkit, RTK Query, Zustand, React Hook Form, D3.js, Highcharts, Framer Motion, Tailwind CSS

Tooling & Practices: Jest, React Testing Library, Webpack, Jenkins, GitHub Actions, Git, Scrum, Design Systems, Figma, WCAG 2.1 AA / Section 508, HIPAA, AI-assisted development (Claude Code, Copilot)